Structured notes: binary operations (Module 1, section 1)

Fraleigh, *A First Course in Abstract Algebra* (8th ed.), Section 1, printed pp. 11–16. Numbers are the book's own. Tables only, no figures in this section. Exercises (printed pp. 17–19) are not included in the notes. Formal version of the same content.

Prerequisites

- $S \times S$ denotes the set of ordered pairs (a, b) with $a, b \in S$ (Definition 0.4).
- Function composition: $(f \circ g)(x) = f(g(x))$.
- Closure of a subset, which is introduced in 1.4 and needed for 1.6–1.7.

1. Binary operations

1.1 Definition

> **DEFINITION 1.1** A binary operation $*$ on a set S is a function from $S \times S$ into S. The value at (a, b) is written $a * b$.

The operation must be defined for every ordered pair of elements of S, and its value must lie in S.

Example 1.2. Addition $+$ and multiplication $\cdot$ are operations on $\mathbb{R}$. Either may be restricted to $\mathbb{C}$, $\mathbb{Z}$, $\mathbb{R}^+$ or $\mathbb{Z}^+$.

Example 1.3. Matrix addition is not an operation on $M(\mathbb{R})$, because $A + B$ is undefined when A and B have different sizes.

1.2 Closure

> **DEFINITION 1.4** Let $*$ be an operation on S and $H \subseteq S$. H is **closed under** $*$ if $a * b \in H$ for all $a, b \in H$. The restriction of $*$ to H is the **induced operation** on H.

Example 1.5. Addition on $\mathbb{R}$ does not induce an operation on $\mathbb{R}^*$: $2, -2 \in \mathbb{R}^*$ but $2 + (-2) = 0 \notin \mathbb{R}^*$.

Example 1.6. Let $H = \{n^2 \mid n \in \mathbb{Z}^+\}$.

- (a) H is not closed under addition: $1, 4 \in H$, but $1 + 4 = 5 \notin H$.
- (b) H is closed under multiplication: $r = n^2$ and $s = m^2$ give $rs = (nm)^2$ with $nm \in \mathbb{Z}^+$.

Example 1.7. Let F be the set of real-valued functions with domain $\mathbb{R}$. For $f, g \in F$ and $x \in \mathbb{R}$, $(f + g)(x) = f(x) + g(x)$, $(f - g)(x) = f(x) - g(x)$, $(f \cdot g)(x) = f(x)g(x)$, and $(f \circ g)(x) = f(g(x))$. F is closed under all four.

1.3 Examples

Example 1.8. On $\mathbb{Z}^+$, $a * b = \min\{a, b\}$. Examples: $2 * 11 = 2$, $15 * 10 = 10$, $3 * 3 = 3$. The operation is commutative.

Example 1.9. On $\mathbb{Z}^+$, $a *' b = a$. Examples: $2 *' 3 = 2$, $25 *' 10 = 25$, $5 *' 5 = 5$. Here $5 *' 7 = 5 \neq 7 = 7 *' 5$, so the operation is not commutative.

Example 1.10. On $\mathbb{Z}^+$, $a *'' b = (a * b) + 2$, where $*$ is as in 1.8. Examples: $4 *'' 7 = 6$, $25 *'' 9 = 11$, $6 *'' 6 = 8$.

Remark. Example 1.8 is used in GPS route selection, where the shorter of two routes is kept.

1.4 Commutativity

> **DEFINITION 1.11** $*$ on S is **commutative** if $a * b = b * a$ for all $a, b \in S$.

Definitions are read as if-and-only-if statements. Theorems are not.

> **PLUS TWO RECALL: COMMUTATIVE LAW**
>
> **Recall.** Addition and multiplication of numbers are commutative.
>
> **Re-derive.**
>
> 1. For $a = 2$, $b = 3$: $2 + 3 = 5 = 3 + 2$.
> 2. For subtraction with the same values: $2 - 3 = -1 \neq 1 = 3 - 2$.
> 3. Commutativity must therefore be verified for each operation.
>
> **Saved.** $*$ of Example 1.8 is commutative; $*'$ of Example 1.9 is not (Definition 1.11).

1.5 Associativity

For three elements there are two bracketings. For the operation of Example 1.8, $(2 * 5) * 9 = 2 = 2 * (5 * 9)$. In general, for that operation,

$$(a * b) * c = a * (b * c).$$

For $*''$ of Example 1.10 the bracketings differ:

$$(2 *'' 5) *'' 9 = 6 \neq 4 = 2 *'' (5 *'' 9).$$

> **DEFINITION 1.12** An operation on S is **associative** if $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$.

Remark. If $*$ is associative, longer products are unambiguous. The book states this without proof.

PLUS TWO RECALL: ASSOCIATIVE LAW

Recall. Addition and multiplication of numbers are associative.

Re-derive.

1. $(2+3)+4 = 9 = 2+(3+4)$.
2. $(2-3)-4 = -5 \neq 3 = 2-(3-4)$.

Saved. Addition and multiplication are associative; subtraction is not (Definition 1.12).

1.6 Associativity of composition

THEOREM 1.13 (Associativity of composition). Let f, g, h map a set S into S. Then $f \circ (g \circ h) = (f \circ g) \circ h$.

Proof. For each $x \in S$, $(f \circ (g \circ h))(x) = f(g(h(x)))$ and $((f \circ g) \circ h)(x) = (f \circ g)(h(x)) = f(g(h(x)))$. The two functions agree at every point. ◆

Remark. Matrix multiplication is associative by this theorem, because it corresponds to composition of linear transformations.

1.7 Identity elements

DEFINITION 1.14 An element $e \in S$ is an **identity element** for $*$ if $a * e = e * a = a$ for all $a \in S$.

Both sides are required in general. If $*$ is commutative, one side suffices.

PLUS TWO RECALL: IDENTITY ELEMENTS

Recall. 0 is the additive identity and 1 the multiplicative identity for real numbers.

Re-derive.

1. $a + 0 = 0 + a = a$: 0 satisfies Definition 1.14 for $+$.
2. $a \cdot 1 = 1 \cdot a = a$: 1 satisfies Definition 1.14 for $\cdot$.
3. For subtraction, $a - 0 = a$, but $0 - a = -a \neq a$ in general: 0 is a right identity but not a left identity.

Saved. 0 and 1 are identities for $+$ and $\cdot$ (Definition 1.14).

THEOREM 1.15 (Uniqueness of identity). A set with a binary operation has at most one identity element.

Proof. Let e and e' be identities. Then $e * e' = e'$ (since e is an identity) and $e * e' = e$ (since e' is an identity). So $e = e'$. ◆

Example 1.16. On F of Example 1.7: $\iota(x) = x$ is the identity for composition; $m(x) = 1$ for multiplication; $a(x) = 0$ for addition. Subtraction of functions has no identity.

1.8 Inverses

DEFINITION 1.17 Let $*$ on S have identity e. For $x \in S$, an **inverse** of x is an element x' with $x * x' = x' * x = e$.

PLUS TWO RECALL: INVERSES

Recall. The additive inverse of a is $-a$; the multiplicative inverse of $a \neq 0$ is $\frac{1}{a}$.

Re-derive.

1. $5 + (-5) = 0$: -5 is an inverse of 5 for $+$.
2. $5 \cdot \frac{1}{5} = 1$: $\frac{1}{5}$ is an inverse of 5 for $\cdot$.
3. 0 has no multiplicative inverse, since $0 \cdot x = 0 \neq 1$.

Saved. Inverses are defined relative to an identity (Definition 1.17).

Example 1.18. On F the operation is composition and the identity is ι. f has an inverse f' with $f \circ f' = f' \circ f = \iota$ exactly when f is one-to-one and onto. This agrees with the usual inverse function.

2. Operations given by tables

A finite operation may be defined by a table. The entry in row i, column j is $(\text{ith entry on the left}) * (\text{jth entry on the top})$.

Example 1.19. Table 1.20 defines $*$ on $\{a, b, c\}$, with $a * b = c$ and $b * a = a$. So $*$ is not commutative.

1.20 Table

$*$	a	b	c
a	b	c	b
b	a	c	b
c	c	b	a

RESULT (TABLE TEST)

A table is commutative if and only if its entries are symmetric about the main diagonal, that is, entry (i, j) equals entry (j, i).

Example 1.21. Complete Table 1.22 so that $*$ is commutative on $\{a, b, c, d\}$. Since $b * a = d$, commutativity forces $a * b = d$. The remaining entries are filled symmetrically, giving Table 1.23.

1.22 Table (given)

$*$	a	b	c	d
a	b			
b	d	a		
c	a	c	d	
d	a	b	b	c

1.23 Table (completed)

$*$	a	b	c	d
a	b	d	a	a
b	d	a	c	b
c	a	c	d	b
d	a	b	b	c

Example 1.24. When an identity exists, it is listed first in the heads, so the first row and column repeat the heads (Table 1.25).

1.25 Table

$*$	e	a	b	c
e	e	a	b	c
a	a	c	c	a
b	b	a	e	c
c	c	e	a	b

3. Conditions for a binary operation

A rule $*$ on S is a binary operation if and only if (1) exactly one element is assigned to each ordered pair, and (2) the element assigned lies in S.

- If (1) fails for some pairs, $*$ is **not everywhere defined**.
- If the assignment can give more than one value, $*$ is **not well defined**.
- If (2) fails, S is **not closed** under $*$.

Example 1.26. $a * b = a/b$: on $\mathbb{Q}$, (1) fails at $b = 0$; on $\mathbb{Q}^*$, both conditions hold; on $\mathbb{Z}^+$, (2) fails, since $1/2 \notin \mathbb{Z}^+$.

Example 1.27. On F, $f * g = f/g$ fails (2): for $f = \cos$ and $g(x) = x^2$, $h(0)$ is undefined, so $h \notin F$.

Example 1.28. On F, $f * g$ is "a function greater than both f and g". This is not well defined: "greater than" is undefined for functions, and in any case many functions qualify.

Example 1.29. On a set of 20 people with distinct heights, $a * b$ is the tallest person. This is a binary operation: it is defined for every pair and gives one person in the set.

Example 1.30. On the same set, $a * b$ is the shortest person taller than both a and b. This is not everywhere defined: if a or b is the tallest person, no such person exists.